Azat Massaurov

Almaty, Kazakhstan · open to relocation to San Francisco

we.just@list.ru · +7 776 207 1122 · Telegram @amassaurov · github.com/massaurov · portfolio

Summary

Systems architect and technical founder with 13+ years building enterprise accounting and production systems in Kazakhstan. Sole owner of a system serving 600 users across 11 branches. I build and run AI agents that act directly on live business databases, and the guardrails that keep them safe. I want to move into full-time empirical research on AI oversight and control.

Selected work

AI agent toolchain for enterprise accounting systems (own project, used in production)

- Retrieval system over 110,000+ procedures of 1C business code (BSL): hybrid search combining BM25 and BGE-m3 embeddings with reciprocal rank fusion, on PostgreSQL/pgvector, served with Python/FastAPI.
- MCP server that lets LLM agents read and act on live 1C databases through COM.
- Guardrails for agent access: data anonymization and protection routines. They contained an incident where an agent tried to clear accounting records as a way to "fix" a problem.

Critic agents for oversight of coding agents

- Separate critic agents review the primary agent's plan and its code before changes are accepted; a written rule defines when each critic runs.

Verified migration of a heavily modified accounting system

- Moved 405 customized objects into an extension with behavior preserved 1:1; comparison with the vendor baseline showed 0 differences.
- Built the verification process: 10 idempotent scripts, reference data sets, a per-object transfer register, timed dry runs with rollback points.
- Full reconciliation caught three critical data-loss cases, including a migration that had silently stopped at 2,000 of 13,389 records.

Experience

1C Architect and Lead Developer, heavy-equipment dealership holding, Kazakhstan (July 2026 – present)

- Sole responsibility for the architecture of the accounting, trade and payroll systems: one database, 11 branches, about 600 users, about 500 documents a day.
- Designed a service-centre system (work orders with a status model, automatic parts reservation, address warehouse with row-level access).
- Migrated 28,000 product items and 1,676 counterparties into production with 0 errors.

Founder and Solution Architect, AZEC, own practice (2019 – present)

- Official 1C franchisee and certified Bitrix24 partner; lead a team of developers.
- Architect and project lead for manufacturing execution systems: a meat-processing plant with 12+ production stages and batch traceability from animal to finished unit; a food factory of about 10,000 m² with 12 workshops.
- Designed a two-tier architecture for a construction holding of 50+ legal entities; took it from specification to a signed contract of 18.5M KZT (200 person-days).
- Automated a production and trading company that moved from accumulated loss to steady profit over a 16-month project.
- Built, automated, brought to profit and sold my own business (a computer club).

1C engineer and administrator, several companies, Kazakhstan (2012 – 2019)

- Implementation and support of accounting and retail systems for small and mid-sized businesses, including administration of 26 databases for a retail chain.

Skills

- **AI and data:** retrieval-augmented generation, hybrid search (BM25, BGE-m3, RRF), pgvector, MCP servers, agent tooling, Anthropic API
- **Engineering:** Python, FastAPI, PostgreSQL, 1C/BSL, REST and COM integrations, data migration and verification
- **Leadership:** system architecture, technical specifications, code review, work with C-level stakeholders

Languages

Russian: fluent. English: beginner (A1).